

Backend Assignment – FastAPI + AI (RAG)

Problem Statement

Build a FastAPI backend that demonstrates good backend development practices while implementing a basic Retrieval-Augmented Generation (RAG) pipeline.

Requirements

- Implement JWT authentication with signup and login endpoints. Passwords must be securely hashed.
- Use either PostgreSQL or MongoDB as the primary database.
- Design appropriate schemas and create suitable indexes. Mention your indexing choices in the README.
- Create an endpoint to ingest text documents. Store the document, chunk it, generate embeddings, and store the chunks and embeddings.
- Implement a /chat endpoint that retrieves relevant chunks and uses an LLM to answer user queries.
- Create custom FastAPI middleware that catches unhandled exceptions and logs them to a PostgreSQL table or MongoDB collection. Log timestamp, endpoint, HTTP method, error message, stack trace, and authenticated user ID (if available). Return a proper JSON error response.
- Organize the project using a clean, modular structure.

Good to Have

- Redis
- Kafka or another message broker
- Docker / Docker Compose
- Background jobs for document ingestion
- Streaming LLM responses
- Unit tests

Submission

Share a GitHub repository with a README containing setup instructions and a .env.example file.

Timeline

- You have a maximum of 1 day (24 hours) to complete this assignment.
- Please push your final code to GitHub and share the repository link before the deadline.

All the Best!